

ASSIGNMENT-1

Deep Learning in Full Stack Web Development

Top 10 Technologies

No	Technology
1	Chatbots
2	Recommendation Systems
3	Image Recognition
4	Sentiment Analysis
5	Fraud Detection
6	Voice Recognition
7	Personalised Content
8	Predictive Analytics
9	Spam Detection
10	Intelligent Search

Project Selected: Image Recognition

Why we chose this project: Image Recognition is one of the most widely used deep learning applications in modern web development. It solves real-world problems by automatically identifying objects, faces and scenes from images. It has applications in healthcare, e-commerce, agriculture, security and smart cities, making it an excellent academic project.

Advantages

- High accuracy in image classification.
- Reduces manual effort through automation.
- Improves user experience with visual search.
- Supports real-time decision making.
- Enhances security using facial recognition.
- Scalable for cloud-based web applications.

Technologies Linked with Deep Learning

Frontend: HTML, CSS, JavaScript, React
Backend: Python, Flask/Django or Node.js
Deep Learning: TensorFlow, Keras, PyTorch, CNN
Database: MySQL or MongoDB
Cloud: AWS, Google Cloud or Azure
Version Control: Git & GitHub

Workflow

User uploads an image → Frontend sends it to the backend → CNN model processes the image → Prediction is generated → Result is displayed on the web application.

Conclusion

Image Recognition combines deep learning with full stack web technologies to build intelligent applications. It offers automation, accuracy and practical value across multiple industries, making it a suitable project for learning and deployment.